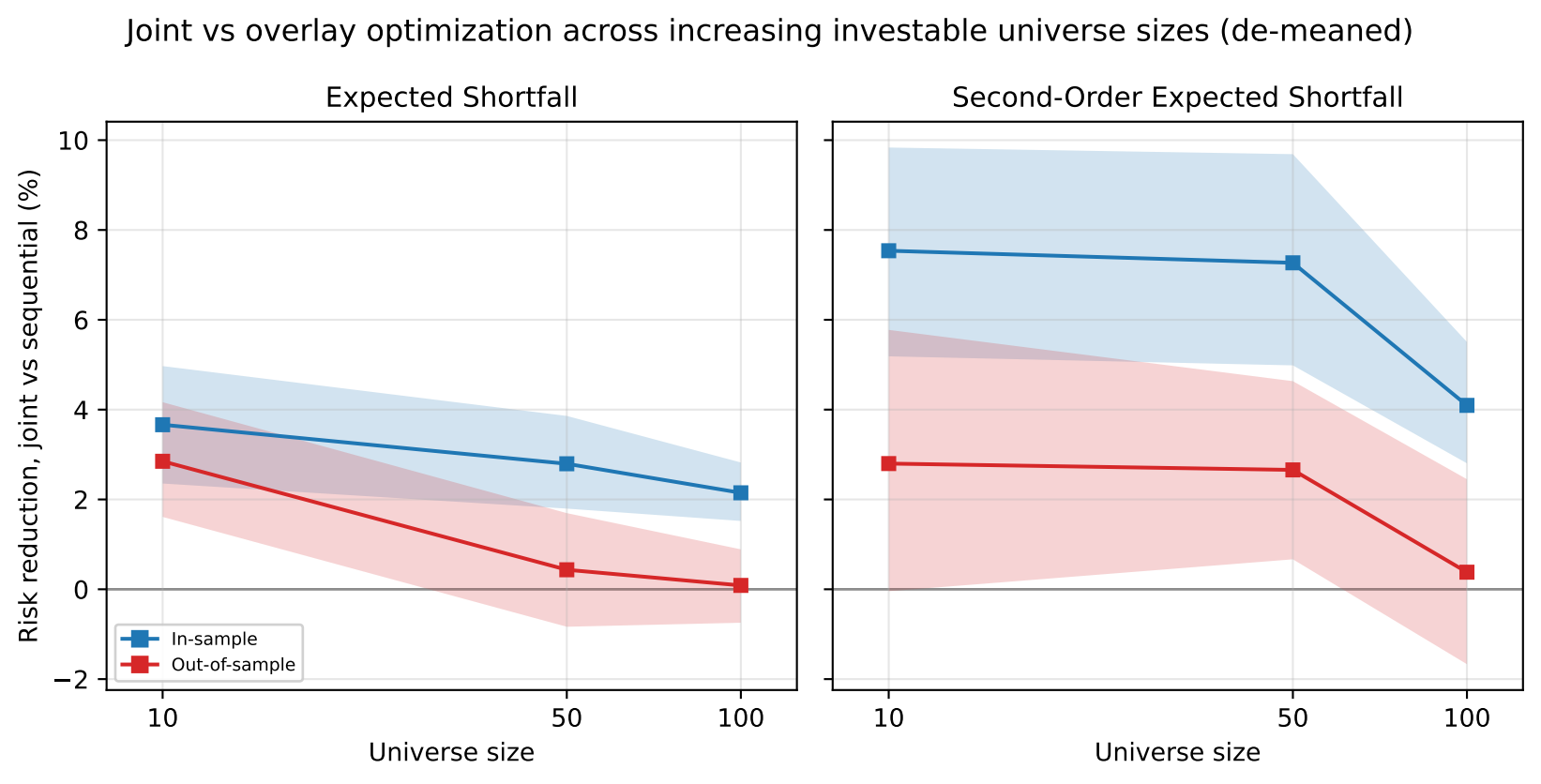
Joint vs overlay optimization across increasing investable universe sizes (de-meaned)
Expected Shortfall
Second-Order Expected Shortfall
Risk reduction, joint vs sequential (%)
10
8
6
4
2
0
−2
In-sample
Out-of-sample
10
50
100
Universe size
10
50
100
Universe size